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Constexpr iterator requirements

I. Introduction and Motivation

This proposal resolves issues [LWG 2897](#) and [LWG 2938](#), which intend to make the iterators of some classes usable in constant expressions. We expect more constexpr usable containers soon from reflection proposals and from work that goes on for an accepted [P0639R0: Changing attack vector of the constexpr vector](#) proposal.

This proposal follows the LWG recommendation and attempts to solve both issues in a generic way by introducing the *constexpr iterator* requirement.

Such requirement will easily allow to make constexpr usable iterators by only adding a few words to the iterator requirements of a container.

Note that swap is excluded from constexpr iterator requirement until [CWG 1581](#) is resolved. Without that resolution making swap constexpr may break user code.

II. Impact on the Standard

This proposal is a pure library extension. It proposes changes that do not break existing code and do not degrade performance. It does not require any changes in the core language.

III. Proposed wording relative to the Working Draft N4687.

Add a new paragraph at the end of 27.2.1 [iterator.requirements.general] as indicated:

-12- An invalid iterator is an iterator that may be singular.(footnote: [...])

Iterators are called *constexpr iterators* if all operations provided to satisfy iterator category operations are constexpr functions, except for

- swap.
- a pseudo-destructor call (8.2.4 [expr.pseudo]), and
- the construction of an iterator with a singular value.

[Note: For example, the types "pointer to int" and reverse_iterator<int*> are constexpr iterators. – end note]

-13- In the following sections, a and b denote [...]

Add a new paragraph at the end of 26.3.7.1 [array.overview]:

-3- An array satisfies all of the requirements of a container [...]

-?- iterator and const iterator satisfy the constexpr iterator requirements (27.2.1 [iterator.requirements.general]).

Add to the end of the 24.4.2 [string.view.template] section:

-1- In every specialization `basic_string_view<charT, traits>`, the type `traits` shall satisfy the character traits requirements (21.2), and the type `traits::char_type` shall name the same type as `charT`.

-?- iterator satisfies the constexpr iterator requirements (27.2.1 [iterator.requirements.general]).

IV. Feature-testing macros

For the purposes of SG10, we recommend updating the values of the feature-testing macros `__cpp_lib_string_view` and `__cpp_lib_array_constexpr` to the date of adoption (expected: 201803).

V. References

[N4687] Working Draft, Standard for Programming Language C++. Available online at www.open-std.org/jtc1/sc22/wg21/docs/papers/2017/n4687.pdf.